

JEE ADVANCED - 9 | PAPER - 1

JEE 2022

Date 27th February, 2022	Maximum Marks 195	Duration 3 Hours
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General Instructions

- The question paper consists of 3 Parts (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **Three** sections (Section 1, Section 2 & Section 3).
- Section 1** contains **10 Multiple Correct Answers Type Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
- Section 2** contains 5 **Multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are Correct.
- Section 3** contains 5 **Numerical Value Type Questions**. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
- For answering a question, an ANSWER SHEET (OMR SHEET) is provided separately. Please fill your **Test Code, Roll No.** and **Group** properly in the space given in the ANSWER SHEET.

Name of the Candidate (In CAPITALS) :

Roll Number :

OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

MARKING SCHEME

SECTION – 1

- This section consists Ten **(10)** Questions. Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four option(s) is correct.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks:	+3 If only (all) the correct option(s) is(are) chosen
Zero Mark:	0 if none of the options is chosen (i.e. the question is unanswered)
Negative Marks:	–1 In all other cases.

SECTION – 2

- This section consists of Five **(05)** Questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks:	+4 If only (all) the correct option(s) is(are) chosen
Partial Marks:	+3 If all the four options are correct but ONLY three options are chosen
Partial Marks:	+2 If three or more options are correct but ONLY two options are chosen and both of which are correct
Partial Marks:	+1 If two or more options are correct but ONLY one option is chosen, and it is a correct option
Zero Mark:	0 if none of the options is chosen (i.e. the question is unanswered)
Negative Marks:	–2 In all other cases.

SECTION 3

- This section has Five **(05)** Questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble 15th $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks:	+3 If ONLY the correct numerical value is entered.
Zero Marks:	0 In all other cases.

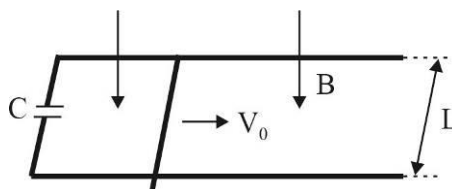
PART I : PHYSICS

65 MARKS

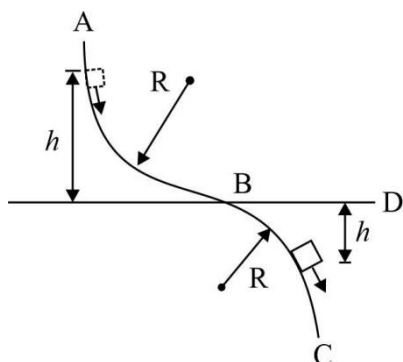
SECTION-1

This section consists of 10 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. Two parallel metal rails, lying at a distance L from each other, are connected by a capacitor of capacitance C at one end. The capacitor is initially uncharged. The arrangement is in a vertical, homogeneous magnetic field B , which is constant in time. A conducting rod of resistance R and mass m is laid perpendicularly onto the rails, and it is given an initial speed V_0 . Determine the final speed of the rod, provided that the rails are long enough, and the homogenous magnetic field extends far enough. (The electric resistance of the rails, the friction and the effects of self induction are negligible.)



- (A) $\frac{V_0}{2}$ (B) $\frac{mV_0}{B^2L^2C}$ (C) $\frac{mV_0}{4B^2L^2C}$ (D) $\frac{mV_0}{m + B^2L^2C}$
2. A thin plano-convex lens acts like a concave mirror of radius of curvature 20 cm when its plane surface is silvered. The radius of curvature of the curved surface if index of refraction of its material is 1.5 will be:
- (A) 40 cm (B) 30 cm (C) 10 cm (D) 20 cm
3. A vernier calliper has its main scale of 10 cm equally divided into 200 equal parts. Its vernier scale of 25 divisions coincides with 12 mm on the main scale. The least count of the instrument is:
- (A) 0.020 cm (B) 0.002 cm (C) 0.010 cm (D) 0.001 cm
4. Figure shows a smooth track in a vertical plane, consisting of two circular arcs AB and BC of the same radius $R = 2m$. The common tangent to the arcs, BD, is horizontal. A small block placed on the track at a height h above BD leaves the track at exactly the same depth below. Determine h .

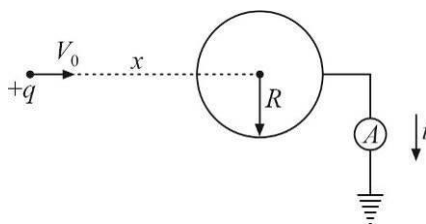


- (A) 10 cm (B) 20 cm (C) 40 cm (D) 80 cm
5. A square loop of wire, edge length a , carries a current i . Compute the magnitude of the magnetic field produced at a point on the axis of the loop at a distance x from the center.

$$a = \sqrt{2}m, x = \sqrt{3}m, i = 1A$$

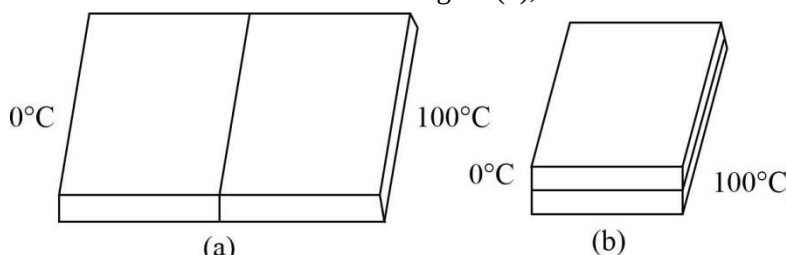
- (A) $\frac{\mu_0}{7\pi}$ (B) $\frac{3\mu_0}{7\pi}$ (C) $\frac{2\mu_0}{7\pi}$ (D) $\frac{5\mu_0}{7\pi}$

6. Find reading of ammeter in given situation. Charge is having velocity V_0 , towards an earthed conducting sphere.



- (A) $\frac{qR}{x^2} V_0$ (B) $\frac{2qR}{x^2} V_0$ (C) $\frac{qx}{R^2} V_0$ (D) $\frac{2qx}{R^2} V_0$

7. Two identical rods of a metal are welded as shown in figure (a). 20 cal of heat flows through them in 4 minute. If the rods are welded as shown in figure (b), then the same amount of heat will flow in:



- (A) 1 minute (B) 2 minute (C) 4 minute (D) 16 minute

8. Light of wavelength $5000\text{\AA}$ and intensity 39.8 W m^{-2} is incident on a metal surface. If only 1% photons of incident light emit photoelectrons, then the number of electrons emitted per second per unit area from the surface will be nearly:

- (A) 10^{18} (B) 10^{22} (C) 10^{24} (D) 10^{26}

9. In a Young's double slit interference experiment, one slit happens to have its width 4 times that of the other. Assuming that intensity of light coming out, is proportional to slit width, the ratio of the maximum to minimum intensity in the interference pattern is:

- (A) 9 (B) $\frac{5}{3}$ (C) 3 (D) $\frac{25}{9}$

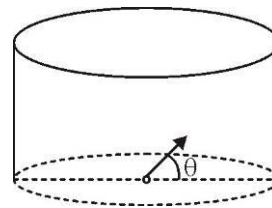
10. The space between two conducting spheres of radii a and b ($a < b$) is filled with a homogenous poorly conducting medium. The capacitance of a such a system equals C . Find the resistivity of the medium if the potential difference between the spheres. When disconnected from an external source decreases n fold during interval τ .

- (A) $\frac{\pi ab\tau}{(b-a)C \log_e n}$ (B) $\frac{4\pi ab\tau}{(b-a)C \log_e n}$
 (C) $\frac{\pi ab\tau}{4(b-a)C \log_e n}$ (D) None of these

SECTION-2

This section consists of 5 Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

11. A particle projected at an angle θ with the horizontal from the centre of the floor of a cylindrical room of radius r returns to the point of projection after three elastic collisions with the walls and the ceiling. If the particle remains in air for time T . The speed of projection.



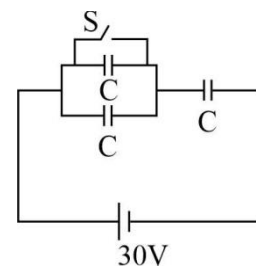
(A) $= \frac{3r}{T \cos \theta}$ (B) $= \frac{4r}{T \cos \theta}$ (C) $= 2\sqrt{\frac{gr}{\sin 2\theta}}$ (D) $> 2\sqrt{\frac{gr}{\sin 2\theta}}$

12. The equation of a wave disturbance is given as $y = 0.02 \sin\left(\frac{\pi}{2} + 50\pi t\right) \cos(10\pi x)$, where x and y are in metres and t is in seconds. Choose the correct statement(s).

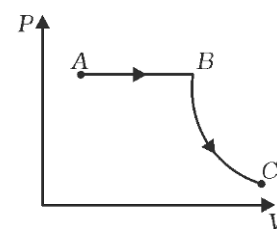
- (A) The wavelength of wave is 0.2 m
 (B) Displacement node occurs at $x = 0.15m$
 (C) Displacement antinode occurs at $x = 0.3m$
 (D) The speed of constituent waves is 0.2 m/s

13. Three capacitors each having capacitance $C = 2\mu F$ are connected with a battery of emf 30V as shown in figure. When the switch S is closed:

- (A) the amount of charge flown through the battery is $20\mu C$
 (B) the heat generated in the circuit is 0.6 mJ
 (C) the energy supplied by the battery is 0.6 mJ
 (D) the amount of charge flown through the switch S is $60\mu C$

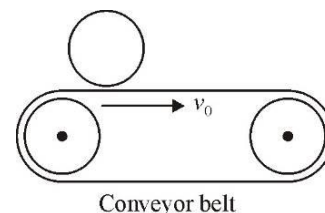


14. A gas with $\gamma = 1.5$ is initially at state A where its pressure, temperature and volume are P_A, T_A and V_A . It is taken first through an isobaric expansion to state $B(P_B, T_B, V_B)$. Then, it is taken through an adiabatic expansion to state $C(P_C, T_C, V_C)$. The work done during the processes AB and BC are equal, i.e. $W_{AB} = W_{BC} = W$. Also, $P_C = \frac{8}{27} P_A$. Then:



(A) $V_C = \frac{27}{4} V_A$ (B) $W = 2P_A V_A$ (C) $V_B = 3V_A$ (D) $\Delta Q_{A \rightarrow B} = 6P_A V_A$

15. A stationary homogenous sphere of radius R and mass M is gently placed on the conveyor belt moving with a constant velocity v_0 towards the east. After a time t from the instant the sphere was placed on the belt, it starts pure rolling and its CM moves with a velocity v relative to the ground. Coefficient of kinetic friction is μ_k .

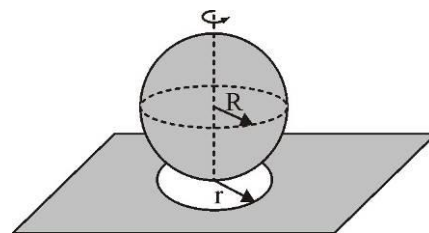


(A) $t = \frac{2v_0}{5\mu_k g}$ (B) $t = \frac{2v_0}{7\mu_k g}$ (C) $v = \frac{2v_0}{5}$ (D) $v = \frac{2v_0}{7}$

SECTION-3

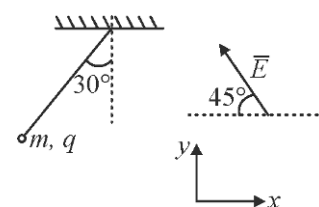
This section consists of **THREE (03)** Paragraphs. There are **TWO (02)** questions corresponding to each Paragraph. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08).

16. A homogeneous sphere of radius $R = 10$ cm rotating about its vertical diameter is gently released on a hole of radius $r_1 = 8.0$ cm made in a fixed horizontal slab. Centre of the hole is vertically below the centre of the sphere. The sphere stops in a time interval $\Delta t_1 = 9.0$ s. Calculate the time interval (in seconds) required to stop the same sphere, if it is placed rotating with the same angular velocity on a hole of radius $r_2 = 6.0$ cm made in another identical horizontal slab.



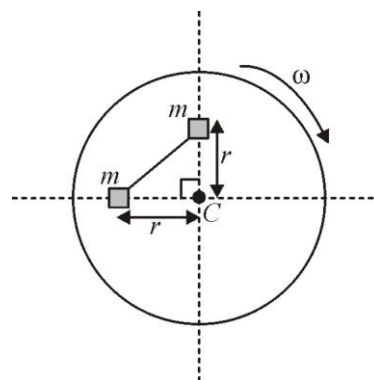
17. Find the quantum number n corresponding to n th excited state of He^+ ion if on transition to the ground state the ion emits two photons in succession with wavelengths 108.5 nm and 30.4 nm. The ionization energy of the hydrogen atom is 13.6 eV.

18. A small ball of mass $m = 55$ gm is hung from a massless thread. Now an electric field $\vec{E} = 10(-\hat{i} + \hat{j}) \text{ N/C}$ is switched on. In equilibrium, the thread makes angle 30° with the vertical. Then, the charge on the ball is _____ C. (Take $g = 10 \text{ m/s}^2$, $\sqrt{3} = \frac{7}{4}$)



19. A force $\vec{F} = 2xy^2 \hat{i} + y \hat{j}$ moves a particle along the curve $y = x^2$ from origin to the point $P(1, 1)$. Work done by the force on the particle from origin to point P is $5/n$ joule. Find n .

20. Two particles of mass m each are kept on a horizontal circular platform on two mutually perpendicular radii at equal distance r from the center of the table. The particles are connected with the string, which is just taut, when the platform is not rotating. The coefficient of static friction between the platform and block is μ . (Now if angular speed of the platform is slowly increased). Find the maximum angular speed ω of platform about its center so that the blocks remain stationary relative to the platform. (If $\mu = \frac{1}{\sqrt{2}}$, $r = 2.5m$, $g = 10 \text{ m/s}^2$)



PART II : CHEMISTRY

65 MARKS

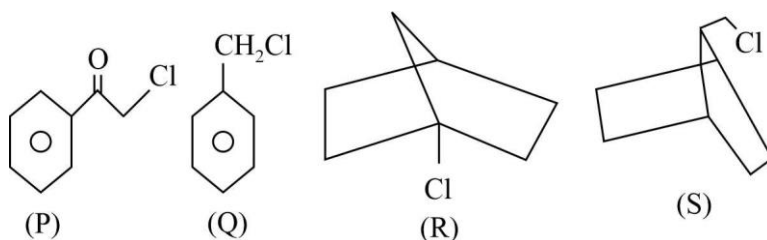
SECTION-1

This section consists of 10 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. In a fuel cell, methanol is used as a fuel and O_2 is used as oxidizer. The standard enthalpy of combustion of methanol is -726 kJ mol^{-1} . The standard free energies of formation of $CH_3OH(l)$, $CO_2(g)$ and $H_2O(l)$ are -166.3 , -394.4 and $-237.1 \text{ kJ mol}^{-1}$ respectively. The standard free energy change of the reaction will be:

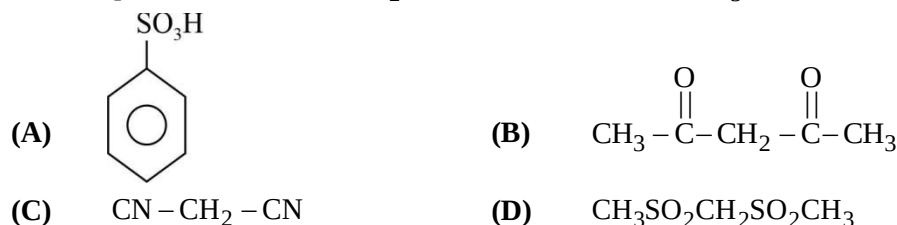
(A) -597.8 kJ (B) -298.9 kJ (C) -465.2 kJ (D) -702.3 kJ

2. $NaBr + H_2SO_4$ undergoes SN_1 reaction with each P, Q, R and S, the rate of reaction vary as :



(A) $P > Q > R > S$ (B) $Q > R > P > S$ (C) $P > R > Q > S$ (D) $Q > S > P > R$

3. The compound that liberate CO_2 on treatment with $NaHCO_3$ is :

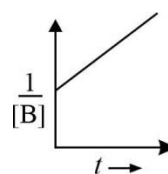


4. Consider the following complex ions P, Q and R. $P = [CoCl_4]^{2-}$, $Q = [Fe(CN)_6]^{4-}$ and $R = [MnCl_4]^{2-}$. The correct order of the complex ions, according to their spin only magnetic values (in B.M) is : (Atomic No. : Mn = 24, Fe = 26, Co = 27)

(A) $R < Q < P$ (B) $Q < R < P$ (C) $R < P < Q$ (D) $Q < P < R$

5. For a reaction $A + B \longrightarrow C$, it is observed that half life of A (when B is taken in very

large amount) is independent of concentration of A and the graph of $\frac{1}{[B]}$ Vs time is



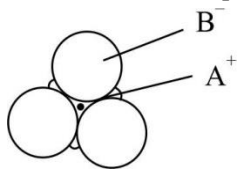
given below when A is in excess. What is order of the reaction ?

(A) 4 (B) 3 (C) 2 (D) 1

6. A pale yellow precipitate and a gas with pungent odour are formed on warming dilute hydrochloric acid with an aqueous solution containing :

(A) sulphate ion (B) sulphide ion
(C) thiosulphate ion (D) sulphite ion

7. The arrangement of B^- ions around A^+ ion in solid AB is given in the figure (not drawn to scale). If the radius of B^- is 180 pm, the radius of A^+ is :



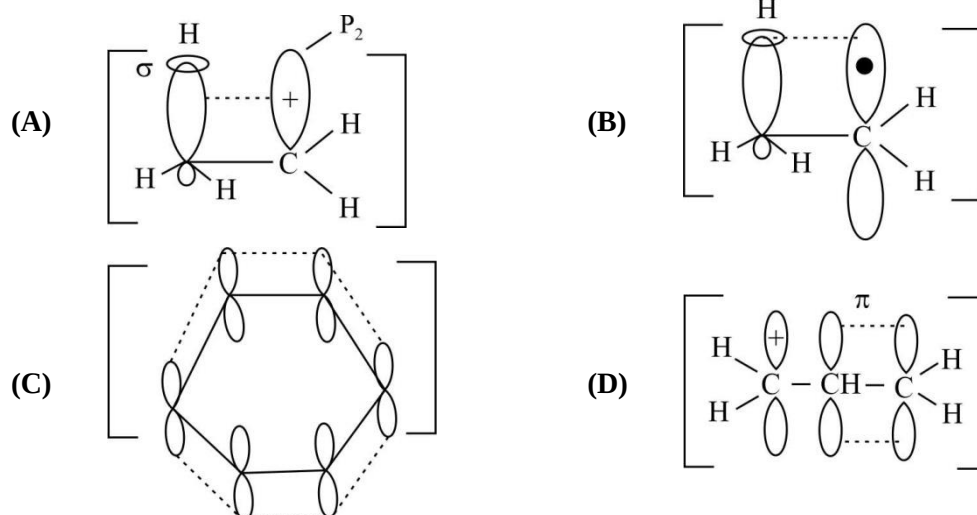
- (A) 41 pm (B) 95 pm (C) 103 pm (D) 110 pm
8. Upon treatment with excess NH_3 solution, the metal ion that will form soluble complex :
- (A) $Cd^{2+}(aq)$ (B) $Mn^{2+}(aq)$ (C) $Pb^{2+}(aq)$ (D) All
9. Although nitrogen does not adsorb on surface at room temperature, it adsorbs on surface at 83K. Which one of the following statements is correct ?
- (A) At 83K, there is formation of monomolecular layer
 (B) At 83K, there is formation of multi molecular layer
 (C) At 83K, nitrogen molecules are held by chemical bonds
 (D) At 83K, nitrogen is adsorbed as atoms
10. Carbonates ores are common for the metals :
- (A) Mg, Zn, Cu (B) Ag, Mg, Cu (C) Na, Mg, Fe (D) Al, Ag, Fe

SECTION-2

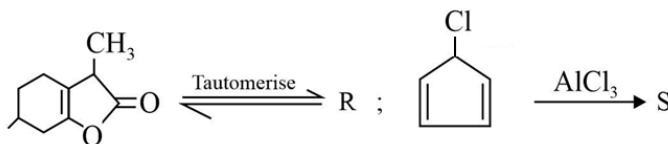
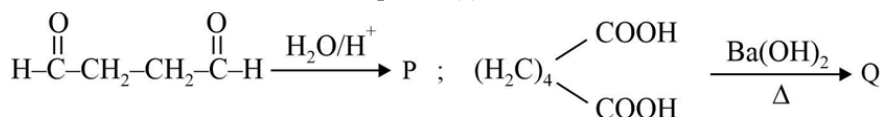
This section consists of 5 Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

11. Carbon tetrachloride and toluene form a non-ideal solution at room temperature: The true statement(s) is/are :
- (A) $\Delta H > 0$ (B) $\Delta S_{\text{system}} > 0$ (C) $\Delta S_{\text{surrounding}} = 0$ (D) ΔG is positive
12. The pair(s) of coordination complexes exhibiting the cis-trans isomerism is/are :
- (A) $[Co(en)_2Cl_2]^+$ and $[Co(en)_2(NH_3)Cl]^{2+}$
 (B) $[Fe(H_2O)_4(OH)_2]^+$ and $[Fe(NH_3)_2(CN)_4]^-$
 (C) $[Cr(Ox)_2Cl_2]^{3-}$ and $[Cr(en)_3]^{3+}$
 (D) $[Co(NH_3)_4(H_2O)Cl]Br$ and $[Co(EDTA)]^{1-}$
13. The dissociation constant of benzoic acid at $25^\circ C$ is 1×10^{-4} . The pH of a 0.01 M solution of its sodium salt is :
- (A) 4 (B) 8 (C) 10 (D) 12

14. Among the following orbital interactions which represents hyper-conjugation interaction ?



15. Among P, Q, R and S the aromatic compound(s) is/are :



- (A) P (B) Q (C) R (D) S

SECTION-3

This section consists of **THREE (03)** Paragraphs. There are **TWO (02)** questions corresponding to each Paragraph. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08).

16. The total number of lone pairs in repeating unit of poly β -hydroxy butyrate-Co- β -hydroxy valerate (PHBV).
17. Oxidation state of central Cr atom in the complex is :
 $[\text{Cr}(\text{O}_2)_2(\text{CN})_3(\text{OH})]$
18. The atomic masses of He and proton are 4 and 1 amu respectively. The value of the de-Broglie wavelength of He gas at -73°C is "M" times that of the de-Broglie wavelength of proton at -223°C . The value of $1/\text{M}$ is :
19. Number of bridging CO ligands in $\text{Fe}_3(\text{CO})_{12}$ is :
20. A tetrapeptide made up of natural amino acid has leucine as the N-terminal residue which is coupled to a chiral amino acid. Upon complete hydrolysis, the tetrapeptide gives alanine, phenylalanine, glycine and leucine. The total number of possible sequences of the tetrapeptide is :

PART III : MATHEMATICS**65 MARKS****SECTION-1**

This section consists of 10 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

- The value of $\tan^{-1}\left(\frac{c_1x-y}{c_1y+x}\right) + \tan^{-1}\left(\frac{c_2-c_1}{1+c_2c_1}\right) + \tan^{-1}\left(\frac{c_3-c_2}{1+c_3c_2}\right) + \dots + \tan^{-1}\left(\frac{c_n-c_{n-1}}{1+c_nc_{n-1}}\right) + \tan^{-1}\left(\frac{1}{c_n}\right)$ is equal to : $\{x, y, c_1, c_2, c_3, \dots, c_{n-1}, c_n > 0\}$
 (A) $\tan^{-1}\left(\frac{y}{x}\right)$ (B) $\tan^{-1}\left(\frac{x}{y}\right)$ (C) $-\tan^{-1}\left(\frac{x}{y}\right)$ (D) None of these
- The points P, Q, R are on the sides OA, OB and OC respectively of a regular tetrahedron $OABC$ are such that $\frac{OP}{OA} = \frac{1}{3}, \frac{OQ}{OB} = \frac{1}{2}, \frac{OR}{OC} = \frac{1}{3}, \frac{Ar(\triangle OAB)}{Ar(\triangle PQR)} =$
 (A) $\frac{9\sqrt{2}}{4}$ (B) $\frac{9\sqrt{2}}{2}$ (C) $\frac{9\sqrt{3}}{2}$ (D) None of these
- The number of points in the complex plane that satisfy the conditions $|z-2|=2, z(1-i) + \bar{z}(1+i) = 4$, (where $i = \sqrt{-1}$) is :
 (A) 0 (B) 1 (C) 2 (D) more than 2
- The base BC of a triangle ABC is bisected at the point (p, q) and the equations to the sides AB and AC are $px + qy = 1$ and $qx + py = 1$. The equations of the median through A is :
 (A) $(p-2q)x + (q-2p)y + 1 = 0$
 (B) $(p+q)(x+y) - 2 = 0$
 (C) $(2pq-1)(px+qy-1) = (p^2+q^2-1)(qx+py-1)$
 (D) $(2pq-3)(px+qy-1) = (p^2+q^2-2)(qx+py-1)$
- The reflection of the point $P(1, 0, 0)$ in the line $\frac{x-1}{2} = \frac{y+1}{-3} = \frac{z+10}{8}$ is :
 (A) $(3, -4, -2)$ (B) $(5, -8, -4)$ (C) $(1, -1, -10)$ (D) $(2, -3, 8)$
- A number is chosen at random from the numbers 10 to 99. By seeing the number a man will laugh if product of the digits is 12. If he chooses three numbers with replacement, then the probability that he will laugh atleast once, is :
 (A) $1 - \left(\frac{31}{45}\right)^3$ (B) $1 - \left(\frac{43}{45}\right)^3$ (C) $1 - \left(\frac{42}{43}\right)^3$ (D) $1 - \left(\frac{41}{45}\right)^3$

7. If $f(x) = \max\left\{\sin x, \cos x, \frac{1}{2}\right\}$, then the area of the region bounded by the curves $y = f(x)$, x -axis, y -axis and $x = 2\pi$ is :
- (A) $\left(\frac{5\pi}{12} + 3\right) \text{sq unit}$ (B) $\left(\frac{5\pi}{12} + \sqrt{2}\right) \text{sq unit}$
 (C) $\left(\frac{5\pi}{12} + \sqrt{3}\right) \text{sq unit}$ (D) $\left(\frac{5\pi}{12} + \sqrt{2} + \sqrt{3}\right) \text{sq unit}$
8. The solution of the differential equation $\{y(1+x^{-1}) + \sin y\}dx + (x + \ln x + x \cos y)dy = 0$ is :
- (A) $x + y \ln x + y \sin x = c$ (B) $xy + y \ln x + x \sin y = c$
 (C) $y + x \ln x + x \sin y = c$ (D) $xy + x \ln x + y \sin y = c$
9. If $f(x) = \int_2^{x^2} \frac{(\sin^{-1} \sqrt{t})^2}{\sqrt{t}} dt$, then the value of $(1-x^2)\{f''(x)\}^2 - 2f'(x)$ at $x = \frac{1}{\sqrt{2}}$ is :
- (A) π^2 (B) $\frac{3\pi}{4}$ (C) $\frac{\pi^2}{4}$ (D) None of these
10. Value of x and y satisfying the equation $\sin^7 y = |x^3 - x^2 - 9x + 9| + |x^3 - x^2 - 4x + 4| + \sec^2 2y + \cos^4 y$ are :
- (A) $x=1, y=n\pi, n \in I$ (B) $x=1, y=2n\pi + \frac{\pi}{2}, n \in I$
 (C) $x=1, y=2n\pi, n \in I$ (D) None of these

SECTION-2

This section consists of 5 Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

11. Let $f(x) = \begin{cases} x^3 + x^2 - 10x & -1 \leq x < 0 \\ \sin x & 0 \leq x < \pi/2 \\ 1 + \cos x & \pi/2 \leq x \leq \pi \end{cases}$ then $f(x)$ has :
- (A) local maxima at $x = \pi/2$ (B) local minima at $x = 0$
 (C) absolute maximum at $x = -1$ (D) absolute maxima at $x = \pi/2$
12. Let $\alpha(n) = 1 - \frac{1}{2} + \frac{1}{3} - \dots + (-1)^{n-1} \frac{1}{n}$, $n \in N$.
- (A) $\frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n} = \alpha(2n)$ (B) $\alpha(2n) < 1 \forall n$
 (C) $\alpha(2n) \geq 0.5 \forall n$ (D) $0.5 < \alpha(n) < 1 \forall n$
13. The point of intersection of the lines $l_1 : \vec{r}(t) = (\hat{i} - 6\hat{j} + 2\hat{k}) + t(\hat{i} + 2\hat{j} + \hat{k})$
 $l_2 : \vec{R}(u) = (4\hat{j} + \hat{k}) + u(2\hat{i} + \hat{j} + 2\hat{k})$
- (A) at $\vec{r}(7)$ (B) at $\vec{R}(4)$ (C) $(8, 8, 9)$ (D) at $\vec{R}(2)$

14. Solutions of the equation $(1 - \tan \theta)(1 + \tan \theta) \sec^2 \theta + 2^{\tan^2 \theta} = 0$ where θ lies in the interval $(-\pi/2, \pi/2)$, given by $\theta_1 < \theta_2 < \dots < \theta_n$, n denotes number of solutions, then :
- (A) $n = 2$ (B) $n = 4$ (C) $\sum_{i=1}^n [\theta_i] = -1$ (D) $\sum_{i=1}^n [\theta_i] = -2$
15. If A and B are two invertible matrices of the same order, then AB is equal to :
- (A) $\frac{1}{|(AB)^{-1}|} \text{adj}((AB)^{-1})$ (B) $|AB| \text{adj}(B^{-1}) \text{adj}(A^{-1})$
- (C) $|AB| \text{adj}(A^{-1}) \text{adj}(B^{-1})$ (D) $\frac{1}{|AB|} \text{adj}(A^{-1}) \text{adj}(B^{-1})$

SECTION-3

This section consists of **THREE (03)** Paragraphs. There are **TWO (02)** questions corresponding to each Paragraph. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08).

16. The equation of the ellipse whose minor axis is equal to the distance between the foci and whose latusrectum is 10 is $mx^2 + ny^2 = 100$, when $m, n \in N$, then the value of $\frac{10^m + 20^n}{82} =$
17. Let $a_r = r^2 \cdot \frac{{}^{100}C_r}{{}^{100}C_{r-1}}$ and λ be the coefficient of x^{99} in $\prod_{i=1}^{100} (x - a_i)$, then the value of $-\frac{\lambda}{55550}$ must be :
18. The position vectors of the points A, B, C and D are : $3\hat{i} - 2\hat{j} - k, 2\hat{i} + 3\hat{j} - 4k, -\hat{i} + \hat{j} + 2k$ and $4\hat{i} + 5\hat{j} + \lambda\hat{k}$. It is known that these points are coplanar, then the value of $-\frac{17\lambda}{146}$ must be :
19. Four squares are chosen at random on a chess board. If the probability that they lie on a diagonal line is $\frac{\lambda}{{}^{64}C_4}$, then the value of $\frac{\lambda}{182}$ must be :
20. Two consecutive numbers from $1, 2, 3, \dots, n$ are removed, then arithmetic mean of the remaining numbers is $\frac{105}{4}$, then $\frac{n}{10}$ must be equal to :